

Task 3 (Group Report) – released by 20/11/2025:

This Task requires you to investigate further the unconstrained and constrained optimisation techniques that have been introduced as part of your Cranfield degree. It is likely to involve some coding, and will involve the production of a detailed report describing your findings.

As part of your coursework for the Computational Optimisation Design module you studied at Cranfield you were required to minimise the Griewank Function by using a stochastic scheme such as Simulated Annealing or Particle Swarm Optimisation. In this and following tasks you are required to investigate techniques of unconstrained and constrained nonlinear optimisation, but you can choose to write code from scratch, use code you have already written, or make use of publicly available source code. The credit you get will be the same, whichever approach you take, but you will, naturally, be expected to acknowledge the authorship of any code you make use of that is not your own. It is important that you have full access to the source code, not 'black box' routines.

Using both the Particle Swarm and the Simulated Annealing techniques, solve three optimisation problems (covering both constrained and unconstrained) of your choice (but NOT related to a cooling tower, and NOT one that you studied for your Computational Optimisation Design coursework) and summarise your results. You are required to compare the performance of these methods. One way of measuring performance is to count the number of evaluations of the objective function on the assumption that, for a suitably 'complicated' objective function, each such evaluation is expensive in terms of time. Use this measure, and any other appropriate measures (e.g. elapsed time), to compare performance. You must include graphical representations of your results.

In this, and subsequent, tasks, if you are employing a penalty function to ensure that constraints are met you should experiment with various multiplying constants to see what effect, if any, they have on ensuring that any constraints are satisfied.

This Task is worth 10% and contributes to Part 1 above.

This report should be sent by email to reach me by midnight 12/12/2026.

Task 4 (Group Report) – released by 11/12/2025:

This Task is similar to Task 3 but involves the investigation of different optimisation techniques.

An alternative, non-stochastic, class of techniques for the numerical solution of (constrained and unconstrained) optimisation is based on an iterative process that at each iteration involves the identification of a search direction and then takes a step in that direction to find a point such that the value of the objective function there is lower than any known so far. Examples are gradient-based methods such as steepest descent, and the Nelder-Mead simplex method. In this Task you are required to investigate an example of this class of solution techniques (henceforth referred to as 'non-stochastic methods'). The technique to be employed should be the quasi-Newton BFGS (Broyden-Fletcher-Goldfarb-Shanno) method. Again, you can develop your own code or make use of existing software. Whether you use your own code, or open-source code, you should carefully match the implementation with the mathematical definition of the method and summarise this in a report (that is, explain how the code implements the mathematical formulation and by so doing make it clear that you understand the mathematics and how the code works).

Using the same optimisation problems as in Task 3, employ the BFGS method to solve them and present your results in numerical and graphical form.

You should prepare a report that summarises your results and compares them with those you obtained in Task 3.

You should additionally provide an agreed (by all group members) indicative set of marks, one for each group member, on a scale of 0-5, that represents the effort of each person in the group. Each grade must be accompanied by a commentary that justifies it.

This Task is worth 10% and contributes to Part 1 above.

This report should be sent by email to reach me by midnight 23/01/2026.

Task 5 (Group Presentation) – released by 23/01/2026:

This Task requires you to present a summary of your findings so far.

During week beginning 26/01/2026 you are required to give a presentation to me, in-person, at ESTIA, on your progress so far. A guideline is 10 slides. You should also hold a group meeting/scrum during this week at which I will be invited to attend as an observer. I will endeavour not to interrupt your discussions unless you specifically ask me for advice.

The presentation part of this Task is worth 10% and contributes to Part 2 above.

You do not need to send anything to me by email in response to this Task.

Task 6 (Group Report) – released by 23/01/2026:

This Task requires you to apply the optimisation techniques employed in Task 3 and Task 4 to a specified optimisation problem.

You are now required to address the design of a hyperboloid cooling tower, and in particular its shape. With this in mind you should regard the cooling tower as a collection of frustums for which the surface of the frustum may be approximated by assuming that it is straight, rather than curved (see Figure 1). You do not need to take into account the width of the wall of the tower.

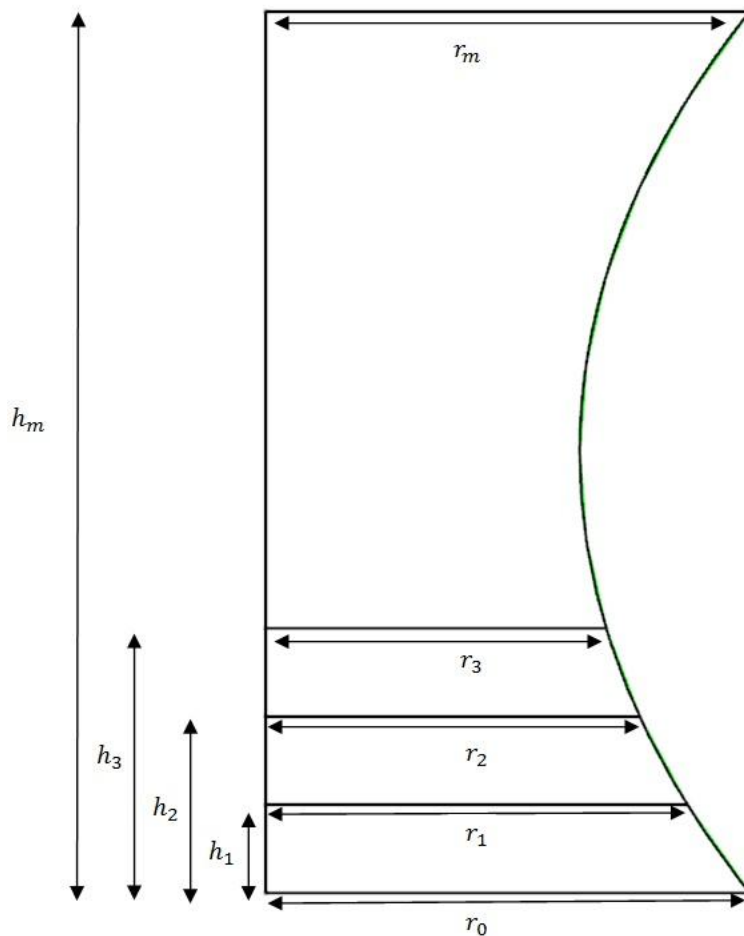


Figure 1 – one half of the cross section of a hyperboloid cooling tower structure regarded as a collection of frustums.

Using this approximation you need to derive formulae for the surface area and volume of each frustum, and hence the volume and surface area of the tower as a whole. (Hint: use the formulae for the surface area and volume of a cone.)

You are now required to use the software that you employed in Tasks 3 and 4 to produce designs for a hyperboloid cooling tower. That is, use two stochastic methods (Particle Swarm and Simulated Annealing), and a non-stochastic method (BFGS). In each case you should minimise the surface area of the tower, and hence construction costs, subject to a fixed volume. The volume itself is open to you to choose to be a realistic value. The parameters that can vary are the radii and the heights of the individual frustums, but you should fix r_0 (base radius) and r_m (top radius). You should consider a variety of situations, for example fix the heights and allow the radii to vary, or fix the radii and allow the heights to vary, or allow both heights and radii to vary. You should also give results for which you have constrained the parameters to lie within given bounds. In all cases you should ensure that the shape of the tower you produce roughly conforms to what you observe in practice, and in particular that it is a hyperboloid. You should consider eight separate (i.e. don't just vary the numbers!) situations. One of the situations you consider should be for a fixed volume of $7.032 \times 10^4 \text{ m}^3$ with $m = 10$, rings r_0 to r_{10} distances 0.0, 3.6, 7.3, 10.9, 14.6, 18.2, 21.9, 25.5, 29.1, 32.8 and 36.5 metres above the base, and r_0 fixed at 39.3 and r_{10} fixed at 27.4.

You should produce images of the cooling towers that your codes produce.

As with Tasks 3 and 4, one aim of this Task is to compare the performance of the numerical techniques that you have employed by, for example, counting the number of function evaluations that are involved, recording the elapsed time, and more.

You should prepare a report that summarises your results and provides commentary on them.